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Unit 9: Rotational Dynamics Lab

Purpose:
To determine the graphical and mathematical relationship between Net Force and Angular Acceleration

Apparatus:

Equipment:

- Spinning apparatus, including a stand, pulley, and spinning mass
- String with an anchor at the end with a changeable mass using washers
- Balance
- Video recording device/iPad

Procedure:

1. The spinning wheel apparatus was set up as in the picture above.
2. The spinning washers were set up by adding washers to each side in order to balance both sides.
3. The length of the string, the radius of the string, and the radius of the disk were measured with a measuring tape, and the mass of a washer was measured with a balance.
4. The time it took for the hanging anchor mass at the bottom of the string from the release of the anchor at its initial position at the top of the apparatus to the bottom of the apparatus was recorded using a video recording device.

5. Eight trials of the time it took for the string to fall was recorded by adding two washers to each previous trial and repeating step 4.
6. The data was recorded in a table on an IPad.

Data:
Table of Mass of anchor in kilograms (kg) vs. Time elapsed in seconds (s)

Mass of anchor (kg)	Time elapsed (s)
0.0375	1.34
0.0628	1.16
0.0881	1.01
0.1134	0.90
0.1387	0.85
0.164	0.74
0.1893	0.68
0.2146	0.62

Analysis:
Table of Net Force in newtons (N) vs Angular Acceleration in radians per second squared (rad/s^2)

Net Force (N)	Angular A... (rad/s^2)
0.375	8.598
0.628	11.473
0.881	15.134
1.134	19.060
1.387	21.368
1.640	28.193
1.893	33.388
2.146	40.163

Derivation of Net Force given the mass of the anchor = m_{anchor} in kilograms (kg) and gravitational constant = g = -10 m/s^2

$$F_{net} = F_G = gm_{anchor}$$

Derivation of Angular Acceleration given the distance traveled by the anchor = Δy in meters (m), radius of the disk = r_D in meters (m), and time elapsed = t in seconds (s)

$$\alpha = \frac{2\Delta y}{r_D t^2}$$

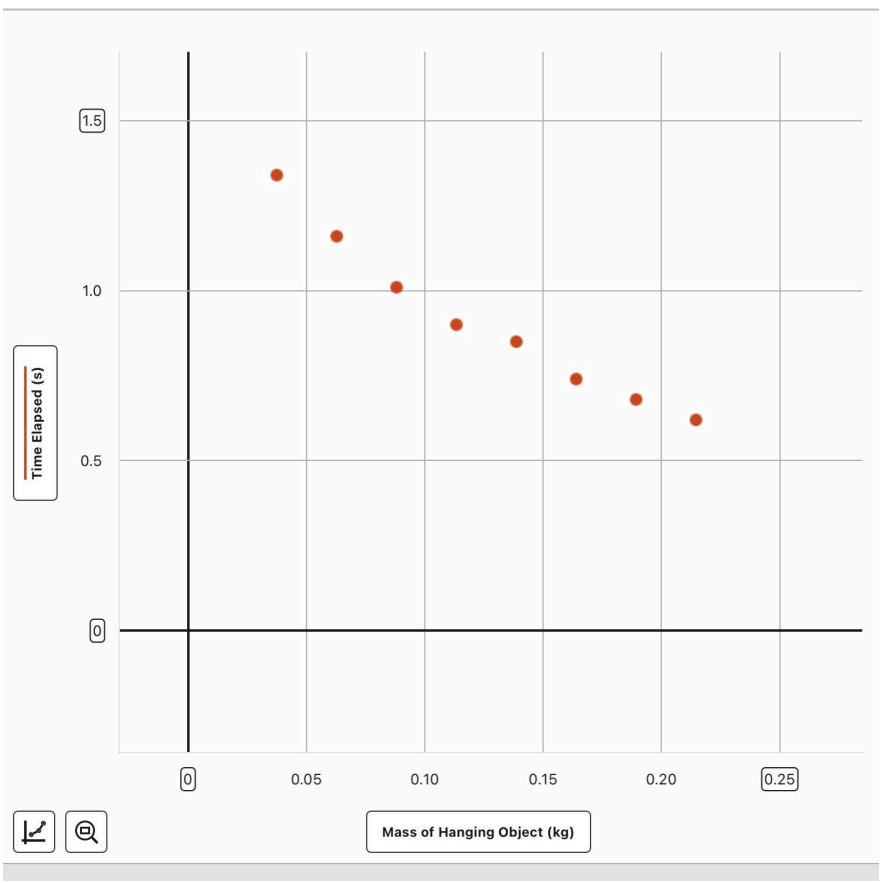
Example calculation of the magnitude of Net Force with Trial 1 shown

$$\|F_{net}\| = \left|(-10(\frac{m}{s^2}))(0.0375(kg))\right| = 0.375(N)$$

Example calculation of Angular Acceleration with Trial 1 shown

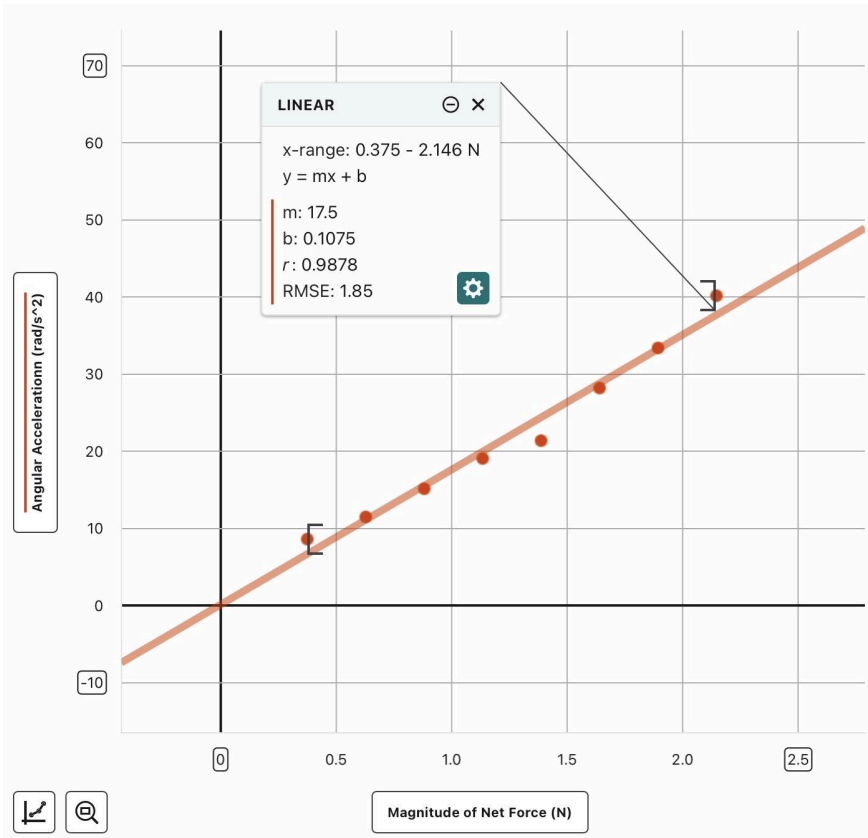
$$\alpha = \frac{2(.88(m))}{(.114(m))(1.34(s))^2} = 8.598(\frac{rad}{s^2})$$

Raw Data Graph of the Mass of the Hanging Object in kilograms (kg) vs. the time elapsed in seconds (s)



Graph of Net Force in Newtons (N) vs. Angular Acceleration in radians per second squared (rad/s^2)

α=(17.5 rad/Ns^2)F_{net} + (0.1075rad/s^2)



Conclusion:
In the final graph of Net Force vs. Angular Acceleration, there appears to be a directly proportional relationship between Net Force and Angular Acceleration. The calculations from the found data using the mass of the anchor and the time elapsed from the release of the string from rest to the time when the anchor hit the bottom of the stand are shown above, as well as the general procedure and logic. The specific linearized equation found from our data is shown below.

$$\alpha = 17.5(\frac{rad}{N \cdot s^2})F_{net} + 0.1075(\frac{rad}{s^2})$$

While the second term suggests the use of a y-intercept representing an initial angular acceleration, the application of the five percent rule is able to be made due to the y-intercept's value of 0.1075 rad/s^2 being less than five percent of the highest vertical axis value of 40.163 rad/s^2, so in reality it is found that the relationship is not simply linear but directly proportional. Of course, an initial angular acceleration (α_0) is possible, but we are assuming that the apparatus is beginning from equilibrium and thereby from zero acceleration.
Thereby, the below statement can be made, relating the net force in Newtons (N) as proportional to angular acceleration in radians per second per second (rad/s^2).

$$\alpha \propto F_{net}$$

In order to make a proportional relationship into an equation, it would be necessary to create a constant *k* that would act as a multiplier for the independent variable, in this case for F_{net}. However, with the findings of the other groups, the relation between angular acceleration = α; the vector distance between the application of the force and the axis of rotation (in this case the radius of the pulley circle) = r_D; the vector net force applied = F_{net}; the mass of the spinning object = *m*; and the distance between the mass of the spinning object to the axis of rotation = r_s is able to be made as seen below. The *k* here as a constant is based on the shape of the object rotating, and by combining the proportionalities of the data across the different lab groups, it would be found that the below expression is directly proportional to the angular acceleration without the need of a *k* or other mathematical constant to multiply the whole expression. The values other than the Net Force were kept constant across this experiment and thereby should be equal or similar to the slope of the line of best fit found in our specific equation. If we were to multiply the radius of the disk = 11.4 cm by the inverse of the mass of the spinning object, approximately (as we did not directly measure it) 0.0506 kg and the squared inverse of the radius of the spinning mass = 7 cm, it should be close to the slope above 17.5. It is not as close as it could be at a value of 4.5979, but this could be attributed to both the approximation of the mass of the spinning rod and the *k* value in the denominator. It is found that this relationship is between an acceleration, a force, and a measure for inertia, which is similar to Newton's Second Law of Motion, so the expression below is known as Newton's Second Law of Rotational Motion, with the numerator equalling the torque = τ and the denominator equalling the Moment of Inertia = *I*. Below the generalized equation is the process of the equating of the slope of the specific equation to the units of the generalized equation, as well as the confirming of the proper units used in the generalized equation.

$$\alpha = \frac{\overrightarrow{r}_D \times \overrightarrow{F}_{net}}{kmr_s^2} = \frac{\overrightarrow{\tau}}{I}$$

$$\frac{rad}{N \cdot s^2} = \frac{1}{N \cdot s^2} = \frac{1}{\frac{kg \cdot m}{s^2} s^2} = \frac{m}{kg \cdot m^2}$$

$$\frac{rad}{s^2} = \frac{1}{s^2} = \frac{m \times \frac{kg \cdot m}{s^2}}{kg \cdot m^2}$$

The use of Rotational Motion necessitates the usage of another model for motion, the rotating bodies model. Instead of seeing objects as one dimensional particles where all forces of an object act upon the same point, we use bodies, collections of infinite points in one object. These bodies take the form of three-dimensional shapes that can rotate, moving the body around an axis of rotation, and translate, changing the spatial coordinates of the whole body without the use of an axis of rotation. Forces can act upon different points of the object, such as gravity acting upon specifically the point of center of mass. Depending on the position of the application of force and the magnitude and direction of the force, the body will rotate or translate.

Some errors that could have occurred in this experiment include the setup of the apparatus. If the spinning masses were not equal in mass or distance from the center of the circle, then imbalances would occur and the data would theoretically fluctuate randomly due to this imbalance. This can be avoided if we take the time to balance the setup of the apparatus. One might have also measured the time elapsed wrongly, the potential lateness of one's ending of the timer leading to a lesser angular acceleration and an earliness leading to a greater angular acceleration. We can avoid this by including more frames per second or including a stopwatch clock on an IPad visible in the video to measure the time elapsed more accurately. The experimenter might also have neglected to carefully let go of the hanging anchor before the timer started, potentially overestimating the angular acceleration if an experimenter tugs on the string down hard enough when they did not tug down on any of the other trials. This can be avoided if we make the apparatus mechanical where we may be pressing a button to let go of the hanging mass, ensuring that this process is constant across trials. Lastly, there might have been nonnegligible friction either between the string and the pulley or the pulley and the stand, causing the time elapsed to be overestimated, underestimating the angular acceleration. The friction from the string and pulley can be reduced by using higher quality materials such as polyester strings. Friction can also be reduced between the pulley and the stand by effectively lubricating the axle and ensuring proper alignment.